

BERT SMS Spam Detection

Model development, evaluation, threshold analysis, and deployment recommendation

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Natural language processing portfolio project

98.21%	93.33%	99.00%	10
Test accuracy	Spam F1	Average Precision	Test errors

Executive Summary

This project evaluates whether a fine-tuned BERT transformer materially improves SMS spam detection over a TF-IDF and Logistic Regression baseline. The final BERT model correctly classified 548 of 558 held-out messages. It achieved 98.21% accuracy, 93.33% spam precision, 93.33% spam recall, a 93.33% spam F1 score, and 99.00% Average Precision.

The main conclusion is measured rather than model-driven: BERT showed excellent probability ranking and contextual capability, but the simpler baseline remained slightly stronger on accuracy, precision, and F1 in this run. Logistic Regression is therefore the preferred low-cost production baseline, with BERT retained as a challenger where contextual understanding justifies additional computing cost.

The model should support a layered filtering workflow. High-confidence spam can be blocked, high-confidence legitimate messages can pass, and uncertain messages can receive secondary screening. Ongoing error monitoring and threshold recalibration remain necessary because spam language changes over time.

Problem Definition and Data

SMS spam detection is a risk-sensitive binary classification task. A false negative allows unwanted or potentially harmful content through the filter. A false positive blocks legitimate communication. Effective evaluation must therefore consider precision, recall, F1, ranking quality, and error types rather than accuracy alone.

Business Question

Can a fine-tuned BERT model improve SMS spam detection while maintaining an acceptable balance between missed spam messages and incorrectly blocked legitimate messages?

Dataset

The analysis uses the UCI SMS Spam Collection through the Hugging Face `ucirvine/sms_spam` dataset. It contains 5,574 English-language SMS messages labeled ham or spam. Stratified sampling preserved the class distribution across training, validation, and testing subsets.

Split	Messages	Ham	Spam	Spam share
Training	4,459	3,861	598	13.41%
Validation	557	483	74	13.29%
Test	558	483	75	13.44%
Total	5,574	4,827	747	13.40%

Evaluation Design

- Training set: model fitting and class-weight estimation.
- Validation set: epoch evaluation and decision-threshold selection.
- Test set: one final estimate of generalization performance.
- Primary class: spam, because missed spam and blocked legitimate messages carry different operational costs.

Modeling Method

Baseline Model

The baseline combines TF-IDF features with Logistic Regression. Unigrams and bigrams represent word occurrence patterns, while balanced class weights reduce the influence of the majority ham class. This model is fast, interpretable, and inexpensive to operate.

Fine-Tuned BERT

The transformer model is google-bert/bert-base-uncased with a sequence-classification head. Messages were tokenized to a maximum length of 128 tokens. Fine-tuning used three epochs, a learning rate of 2e-5, batch size 16, weight decay of 0.01, a linear learning-rate schedule, gradient clipping, and 84 warm-up steps.

Class Imbalance Treatment

Training contained 3,861 ham messages and 598 spam messages. Weighted cross-entropy assigned approximate class weights of 0.5774 to ham and 3.7283 to spam. This increased the penalty for missing the minority spam class.

Evaluation Metrics

Metric	Purpose
Accuracy	Share of all messages classified correctly
Precision	Share of blocked messages that were truly spam
Recall	Share of available spam successfully detected
F1 score	Balance between spam precision and recall
ROC-AUC	Ranking quality across false-positive-rate thresholds
Average Precision	Precision-recall ranking quality for the minority spam class

Reproducibility Controls

The data split used a fixed random seed and stratification. Threshold selection occurred only on validation data. Training results may still vary slightly because transformer initialization and some GPU operations are nondeterministic.

Results

The final BERT model produced balanced spam precision and recall. The baseline remained extremely competitive and slightly exceeded BERT on accuracy, precision, and F1, while BERT increased recall relative to the baseline.

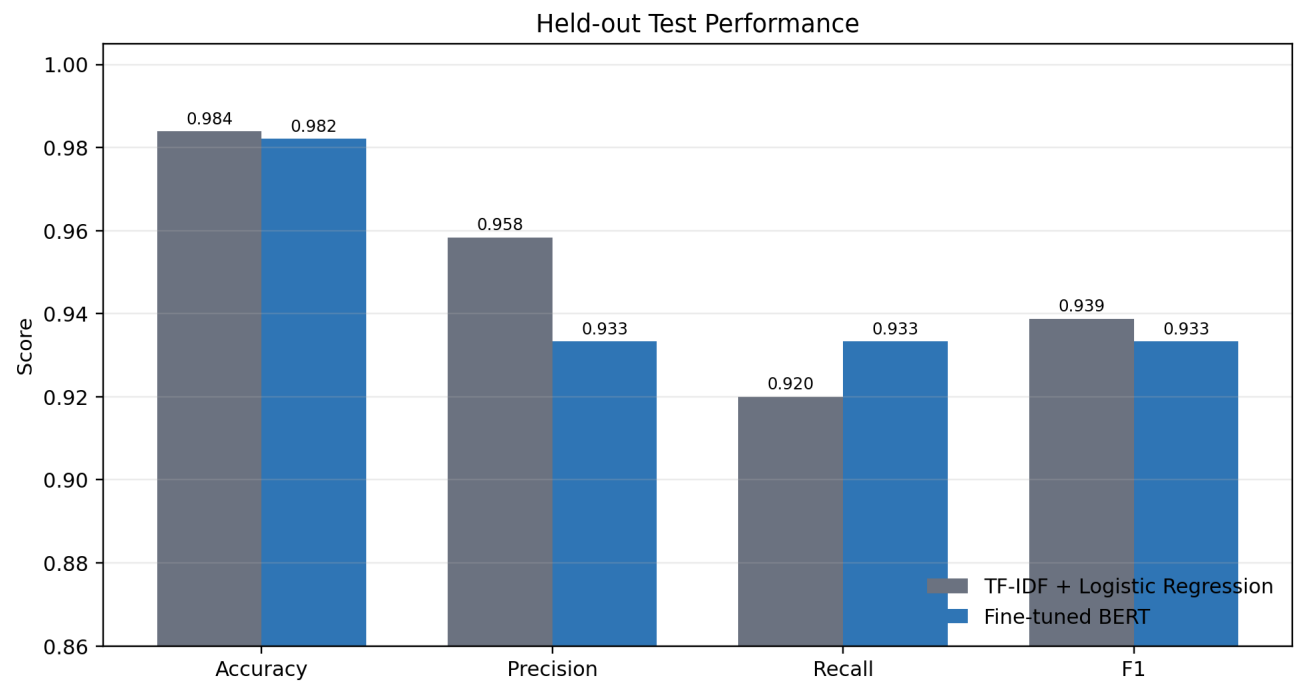


Figure 1. Held-out performance of the TF-IDF baseline and fine-tuned BERT.

Model	Accuracy	Precision	Recall	F1
TF-IDF + Logistic Regression	98.39%	95.83%	92.00%	93.88%
Fine-tuned BERT	98.21%	93.33%	93.33%	93.33%

Interpretation

BERT detected one additional share of spam relative to the baseline, but it also generated more false alerts. The difference in F1 was small. BERT's 99.00% Average Precision shows strong ranking capability, yet that strength did not translate into a clear default-threshold advantage over the simpler model in this test sample.

Threshold and Error Analysis

Validation Threshold Selection

Candidate thresholds from 0.10 to 0.90 were evaluated on the validation set. Multiple thresholds tied for the maximum F1 score. The tie was resolved by selecting the candidate closest to the conventional 0.50 cutoff, avoiding an arbitrary preference for the first tied value.

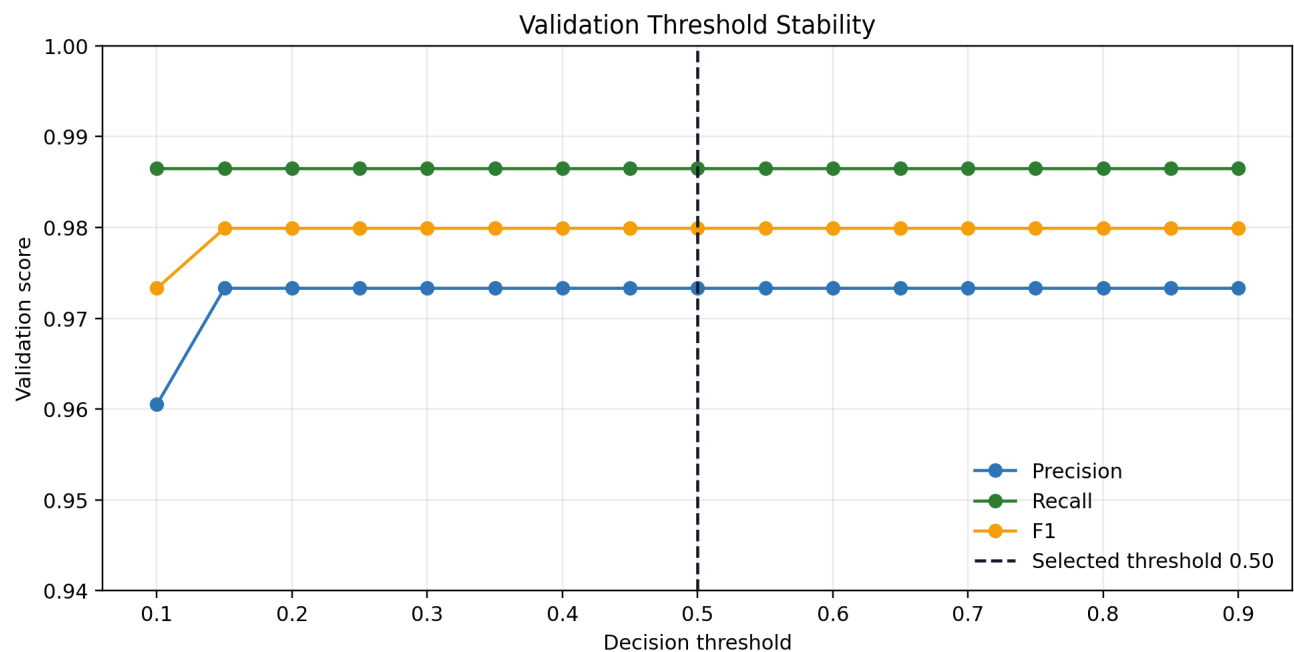


Figure 2. Validation metrics remained stable across a broad threshold range.

Selected threshold	Precision	Recall	F1	False positives	False negatives
0.50	97.33%	98.65%	97.99%	2	1

Threshold stability indicates that most validation probabilities were well separated. It does not prove perfect calibration, and it should be reassessed on newer messages and under changing spam tactics.

Test Errors and Operational Risk

At the selected threshold, BERT made 10 errors on 558 held-out messages. Five legitimate messages were incorrectly flagged, and five spam messages were missed.

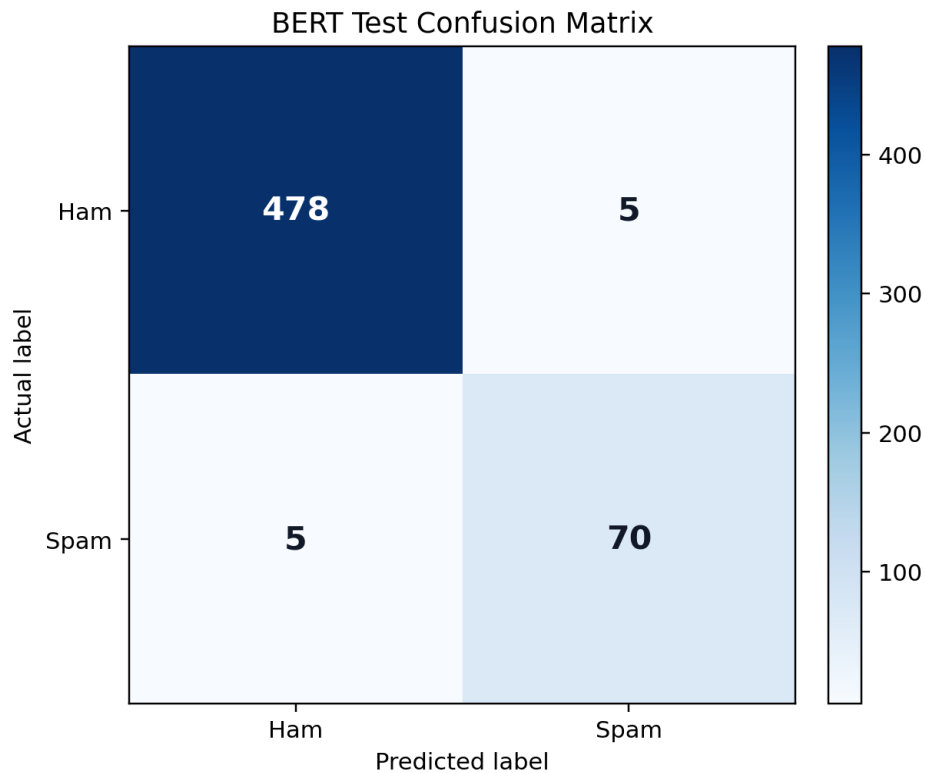


Figure 3. The held-out confusion matrix contains 478 true ham, 70 true spam, 5 false positives, and 5 false negatives.

Error Implications

- False positives can delay or block legitimate personal and business communication.
- False negatives can expose recipients to promotions, phishing attempts, or fraud-related language.
- Short messages, abbreviations, phone numbers, and missing conversational context remain difficult signals.
- Probability scores should be logged so analysts can identify drift, repeated error patterns, and threshold changes.

Recommendation and Limitations

Model Recommendation

Use TF-IDF with Logistic Regression as the initial production baseline when inference speed, operating cost, and interpretability are the leading constraints. Retain BERT as a challenger model for controlled testing, especially when new data show that contextual wording creates errors the baseline cannot resolve.

Deployment Design

- Block only high-confidence spam automatically.
- Pass high-confidence ham without intervention.
- Send uncertain messages to a second-stage rule, model, or review queue.
- Track false-positive and false-negative rates by time period and message type.
- Recalibrate the decision threshold when business costs or message patterns change.
- Retrain and retest when language drift or new spam campaigns reduce performance.

Limitations

- The dataset contains 5,574 messages and may not represent current multilingual, multimedia, or adversarial spam.
- The project compares model quality but does not benchmark inference latency, memory use, or monetary operating cost.
- Average Precision measures ranking quality; calibration should be assessed separately before probability-based actions.
- The demonstration is a decision-support prototype, not a production messaging-security system.

Conclusion

The fine-tuned BERT model delivered strong and balanced spam detection, but the experiment did not justify replacing the simpler baseline automatically. The strongest outcome is the decision framework: compare models fairly, tune thresholds without test leakage, inspect the costs of both error types, and select the deployment architecture that matches operational priorities.

References

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